

An internal controlled wave problem: Schwarz and Dirichlet-Neumann parallel in time methods

Thanh Vuong DANG

LAGA, Ecole doctorale Galilée, Université Sorbonne Paris Nord

ANR Allowap

Optimal control problem

MAIN PROBLEM

Optimal control problem

We minimize the functional $\hat{J}$.

$$\min_u \hat{J}(u) = J(u, y),$$

$$J(u, y) = \frac{1}{2} \int_{\Omega \times [0, T]} |y(x, t) - \hat{y}(x, t)|^2 dx dt \\ + \frac{\gamma}{2} \int_{\Omega} |y(x, T) - \hat{z}(x)|^2 dx + \frac{\alpha}{2} \int_{\Omega \times [0, T]} |u(x, t)|^2 dx dt,$$

where $u \in L^2(\Omega \times (0, T))$ is a control and y is the solution

$$\begin{cases} y_{tt} - \Delta y = u, & (x, t) \in \Omega \times [0, T], \\ y(x, t) = 0, & x \in \partial\Omega, t \in [0, T], \\ y(x, 0) = y^0(x), y_t(x, 0) = y^1(x), & x \in \Omega, \end{cases}$$

where $(y^0(x), y^1(x)) \in L^2(\Omega) \times H^{-1}(\Omega)$.

Well-posedness

Minimization problem

$$\begin{aligned} \min_{u \in L^2(\Omega)} \hat{J}(u) = J(u, y) &= \frac{1}{2} \int_{\Omega \times [0, T]} |y(x, t) - \hat{y}(x, t)|^2 dx dt \\ &+ \frac{\gamma}{2} \int_{\Omega} |y(x, T) - \hat{z}(x)|^2 dx + \frac{\alpha}{2} \int_{\Omega \times [0, T]} |u(x, t)|^2 dx dt, \end{aligned}$$

Theorem (1)

The minimization problem is well-posed.

Proof of well-posedness

Using formula Taylor-Young, we have

$$\hat{J}''(u)(v, v) \geq \alpha \int_{\Omega \times (0, T)} |v(x, t)|^2 dx dt.$$

$\hat{J}$ is elliptic then $\hat{J}$ admits a unique minimizer u .

Optimality system

Optimality system

$$\left\{ \begin{array}{ll} y_{tt} - \Delta y = \frac{\lambda}{\alpha}, & \text{on } \Omega \times (0, T) \\ \lambda_{tt} - \Delta \lambda + y - \hat{y} = 0, & \text{on } \Omega \times (0, T) \\ \lambda(., T) = 0, & \text{on } \Omega \\ \gamma(y(., T) - \hat{z}) - \lambda_t(., T) = 0, & \text{on } \Omega \\ \lambda = \alpha u, & \text{on } \Omega \times (0, T) \\ \lambda = 0, & \text{on } \partial\Omega, \\ y = 0, & \text{on } \partial\Omega \times [0, T], \\ y(x, 0) = y^0(x), y_t(x, 0) = y^1(x), & \text{on } \Omega. \end{array} \right.$$

Discretization

We discretize the system by using the finite volume method (see [Gander, Halpern, Nataf]) and we get the convergence order is h^2 ($h = dx$).

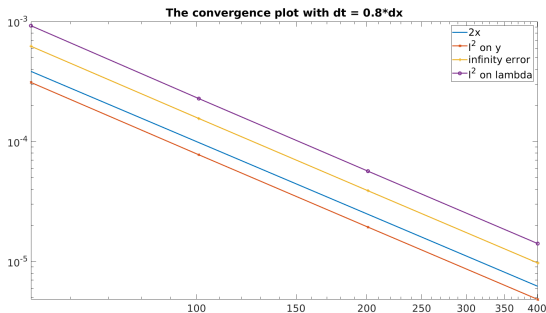


Figure: Convergence plot of order h^2

SCHWARZ METHOD IN TIME

Bibliography



Matthias Heinkenschloss, A time-domain decomposition iterative method for the solution of distributed linear quadratic optimal control problems, 2003.



Felix Kwok, On the Time-domain Decomposition of Parabolic Optimal Control Problems, 2017.



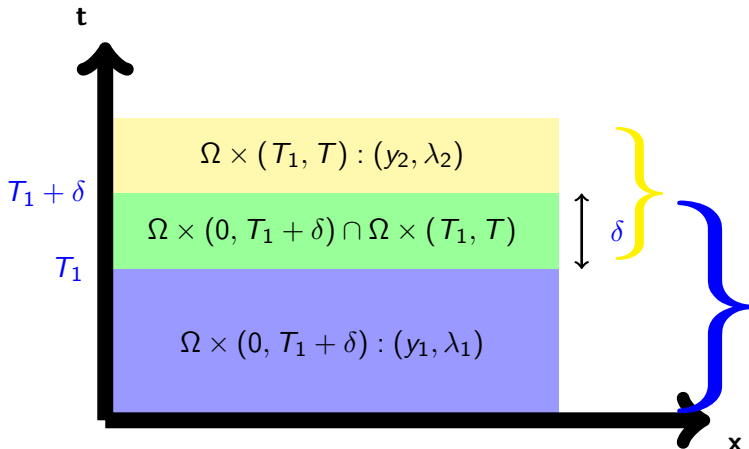
Martin J.Gander and Felix Kwok, Schwarz Methods for the Time-Parallel Solution of Parabolic Control Problems, 2016.



J.E.Lagnese and G.Leugering, Time domain decomposition of optimal control problems for the wave equation, 2002.

Domain decomposition

We consider two subdomains of $\Omega \times (0, T)$, which are $\Omega \times (0, T_1 + \delta)$ and $\Omega \times (T_1, T)$.



Time domain decomposition algorithm.

$$\begin{cases} y_{1,tt}^{n+1} - y_{1,xx}^{n+1} - \frac{1}{\alpha} \lambda_1^{n+1} = 0, & (x, t) \in \Omega \times (0, T_1 + \delta), \\ \lambda_{1,tt}^{n+1} - \lambda_{1,xx}^{n+1} + y_1^{n+1} = \hat{y}, & (x, t) \in \Omega \times (0, T_1 + \delta), \\ y_1^{n+1}(x, 0) = y_{1,t}^{n+1}(x, 0) = 0, & x \in \Omega, \\ \lambda_1^{n+1}(x, T_1 + \delta) = \lambda_2^n(x, T_1 + \delta), & x \in \Omega, \\ \lambda_{1,t}^{n+1}(x, T_1 + \delta) - \gamma y_1^{n+1}(x, T_1 + \delta) = \\ \lambda_{2,t}^n(x, T_1 + \delta) - \gamma y_2^n(x, T_1 + \delta), & x \in \Omega, \end{cases}$$

$$\begin{cases} y_{2,tt}^{n+1} - y_{2,xx}^{n+1} - \frac{1}{\alpha} \lambda_2^{n+1} = 0, & (x, t) \in \Omega \times (T_1, T), \\ \lambda_{2,tt}^{n+1} - \lambda_{2,xx}^{n+1} + y_2^{n+1} = \hat{y}, & (x, t) \in \Omega \times (T_1, T), \\ \lambda_2^{n+1}(x, T) = 0, & x \in \Omega \\ \lambda_{2,t}^{n+1}(x, T) = \gamma(y_2^{n+1}(x, T) - \hat{z}), & x \in \Omega \\ y_2^{n+1}(x, T_1) = y_1^{n+1}(x, T_1), & x \in \Omega, \\ y_{2,t}^{n+1}(x, T_1) = y_{1,t}^{n+1}(x, T_1), & x \in \Omega. \end{cases}$$

Well-posedness

Theorem (3)

Time domain system has unique solution for each iteration.

Sketch of proof:

- Write the minimization problem for each domain 1 and 2.
- Utilize alpha convexity to prove the uniqueness solution on each domain for corresponding iteration.

ANALYSIS OF TOY MODEL

Schwarz algorithm

In our case, we take $\alpha = 1, \gamma = 0, \delta = 0$.

$$\begin{cases} y_{1,tt}^{n+1} - y_{1,xx}^{n+1} - \lambda_1^{n+1} = 0, & (x, t) \in \Omega \times (0, T_1), \\ \lambda_{1,tt}^{n+1} - \lambda_{1,xx}^{n+1} + y_1^{n+1} = \hat{y}, & (x, t) \in \Omega \times (0, T_1), \\ y_1^{n+1}(x, 0) = y_{1,t}^{n+1}(x, 0) = 0, & x \in \Omega, \\ \lambda_1^{n+1}(x, T_1) = \lambda_2^n(x, T_1), & x \in \Omega, \\ \lambda_{1,t}^{n+1}(x, T_1) = \lambda_{2,t}^n(x, T_1), & x \in \Omega, \end{cases}$$

$$\begin{cases} y_{2,tt}^{n+1} - y_{2,xx}^{n+1} - \lambda_2^{n+1} = 0, & (x, t) \in \Omega \times (T_1, T), \\ \lambda_{2,tt}^{n+1} - \lambda_{2,xx}^{n+1} + y_2^{n+1} = \hat{y}, & (x, t) \in \Omega \times (T_1, T), \\ \lambda_2^{n+1}(x, T) = \lambda_{2,t}^{n+1}(x, T) = 0, & x \in \Omega \\ y_2^{n+1}(x, T_1) = y_1^{n+1}(x, T_1), & x \in \Omega, \\ y_{2,t}^{n+1}(x, T_1) = y_{1,t}^{n+1}(x, T_1), & x \in \Omega. \end{cases}$$

Theoretical convergence factor

- We make an assumption that Ω is infinite domain.

- We apply Fourier transform in space. Let $(\lambda_0)^n = (\hat{\lambda}_1^n(k, 0))$, $(\lambda'_0)^n = (\hat{\lambda}_{1,t}^n(k, 0))$ and $(y_T)^n = (\hat{y}_2^n(k, T))$, $(y'_T)^n = (\hat{y}_{2,t}^n(k, T))$.

- Then,

$$\begin{pmatrix} (\lambda_0)^{n+1} \\ (\lambda'_0)^{n+1} \end{pmatrix} = M(k) \begin{pmatrix} (\lambda_0)^n \\ (\lambda'_0)^n \end{pmatrix}. \quad (1)$$

- The convergence factor is the spectral radius of the matrix $M(k)$ for each frequency k .

Schwarz algorithm

Convergence of toy model of Schwarz algorithm.

Theorem (4)

Given $\alpha = 1, \delta = 0, \gamma = 0$, there exists a $\tilde{T}_0$ such that for $T < \tilde{T}_0$, $T_1 = \frac{T}{2}$ the convergence factor $\rho^ = \max_{[0, k_{\max}]} \rho(M(k)) < 1$.*

Proof

- Bound the $\rho(M) \leq \|M\|_\infty < c(T_1) < 1$, where

$$c(T_1) = \frac{1}{\sqrt{2}} \sinh(T_1 \sqrt{2}) + \frac{1}{2} \left(\cosh \left(\frac{1}{\sqrt{2}} T_1 \right)^2 - 1 \right) \\ + \left(\sin \left(\frac{1}{\sqrt{2}} T_1 \right) \right)^2 + \frac{\sqrt{2} \cosh(T_1 \sqrt{2}) |\sin(T_1 \sqrt{2})|}{4} \\ = \max\{c^+(T_1), c^-(T_1)\},$$

- $c^+(T_1)$ is increasing and continuous function with $c^+(0) = 0$ and $\lim_{T_1 \rightarrow \infty} c^+(T_1) = \infty$.
- There exists a $\tilde{T}_1^+$ such that $c^+(\tilde{T}_1^+) = 1$.
- Similarly, there exists a $\tilde{T}_1^-$ such that $c^-(\tilde{T}_1^-) = 1$.
- Choose $\tilde{T}_0 = 2\tilde{T}_1 = 2 \min\{\tilde{T}_1^+, \tilde{T}_1^-\}$, which is the point satisfies the theorem.

Numerical illustration for function $c(T_1)$

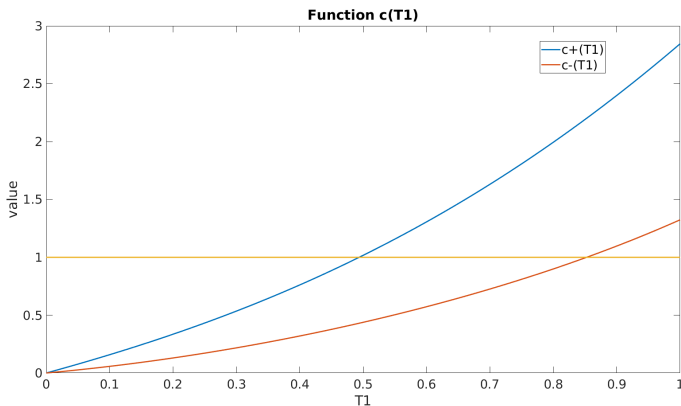


Figure: Function $c(T_1)$

Numerical observation

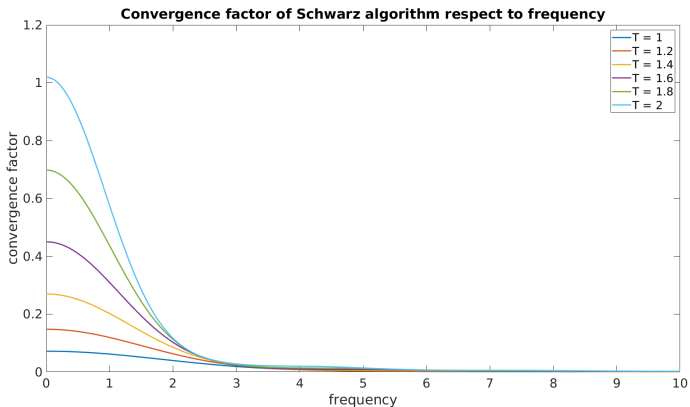


Figure: Convergence factor of Schwarz method

Schwarz algorithm

Corollary (1)

Let $T^* \leq \tilde{T}_0$ and $T_1 = \frac{T}{2}$. There is a neighborhood of $(T^*, \alpha^*, \gamma^*, \delta^*) = (T^*, 1, 0, 0)$ such that $\rho^*(T, \alpha, \gamma, \delta) = \max_{[0, k_{\max}]} \rho(M(k, T, \alpha, \gamma, \delta)) < 1$.

Sketch of proof:

- We know that the coefficient of matrix M is continuous respect to $(T, \alpha, \gamma, \delta)$.
- Utilize the definition of continuous function.

Schwarz algorithm

Theorem (5)

There exists a $T_0 > 0$ such that for

$T > T_0$, $T_1 = \frac{T}{2}$, $\alpha = 1$, $\gamma = 0$, $\delta = 0$, the convergence factor

$\rho^ = \max_{[0, k_{\max}]} \rho(k) > 1$.*

Proof of theorem

- Asymptotic analysis for variable T_1 for $k = 0$.
- Characteristic equation: $\lambda^2 - \text{trace}(M)\lambda + \det(M) = 0$.
- We compute $\lim_{T_1 \rightarrow \infty} \det(M) = 1$ and $\lim_{T_1 \rightarrow \infty} \text{trace}(M) = -6$.
- We have the characteristic equation $\lambda^2 + 6\lambda + 1 = 0$. One of root is bigger than 1.

Schwarz algorithm for T large

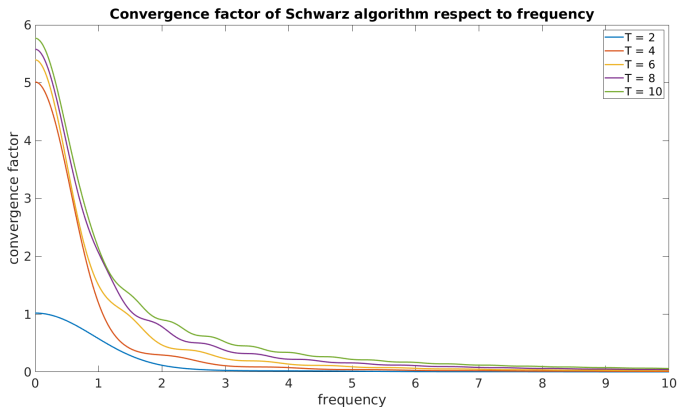


Figure: Convergence factor of Schwarz method when T large

Schwarz algorithm for T large (numerics)

Iterative error with zero solution with $T1 = T/2$, $\alpha = 1$, $\delta = 0$, $N=51$

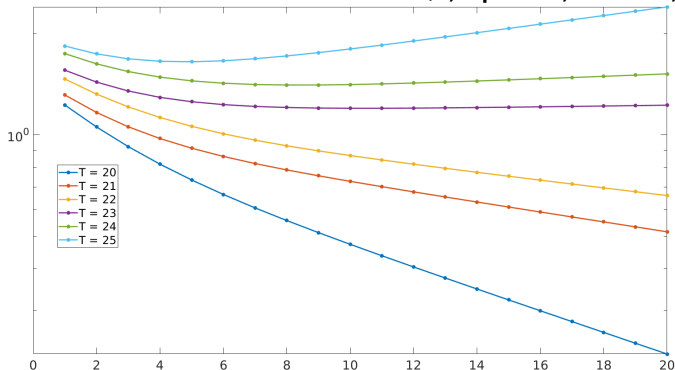


Figure: Iterative error when T large

Optimized Schwarz algorithm

- Changing the transmission condition for domain 1 with parameter θ , we have Schwarz relaxation algorithm.

$$\begin{cases} \lambda_1^{n+1}(x, T_1) = (1 - \theta)\lambda_1^n(x, T_1) + \theta\lambda_2^n(T - T_1), \\ \lambda_{1,t}^{n+1}(x, T_1) = (1 - \theta)\lambda_{1,t}^n(x, T_1) + \theta\lambda_{2,t}^n(T - T_1). \end{cases}$$

- Convergence factor is spectral radius of

$$M_\theta(k) = I - \Theta(I - M(k)).$$

Conjecture (1)

Given $T > 0$, there exists some values of $\theta \in [0, 1]$ such that Schwarz algorithm converges.

Estimate the optimal theta

Theorem (6)

Let $\theta_{opt} = \min_{\theta \in [0,1]} \max_{[0, k_{max}]} \rho(M_{\theta}(k))$, when $T \rightarrow +\infty$,
 $\theta_{opt}(T) \rightarrow \theta_0 = 0.25$ and $\rho_{opt}(T) \rightarrow \frac{1}{\sqrt{2}}$.

Proof of theorem

- We use the asymptotic expansion for M when T_1 large.
- From here we have the eigenvalue of M respect to k

$$\begin{cases} s_1 \sim - \left(-\frac{1}{2\nu_1^2} + \frac{|\nu|}{\nu_1} \right)^2 + O(\exp(-2\nu_1 T_1)), \\ s_2 \sim - \left(-\frac{1}{2\nu_1^2} - \frac{|\nu|}{\nu_1} \right)^2 + O(\exp(-2\nu_1 T_1)), \end{cases}$$

where $(\nu_1 + i\nu_2)^2 = k^2 + i$ and $\nu = (\nu_1, \nu_2)$.

Minimax problem

- To find the limit of optimal θ , we have to solve the min max problem (Here we consider s_1 as $\lim_{T_1 \rightarrow +\infty} s_1$ same with s_2).

$$\min_{\theta \in [0,1]} \max_{k \in [0, k_{max}]} \max\{|1 + \theta(s_1 - 1)|, |1 + \theta(s_2 - 1)|\}$$

Asymptotic convergence factor of Schwarz respect to theta

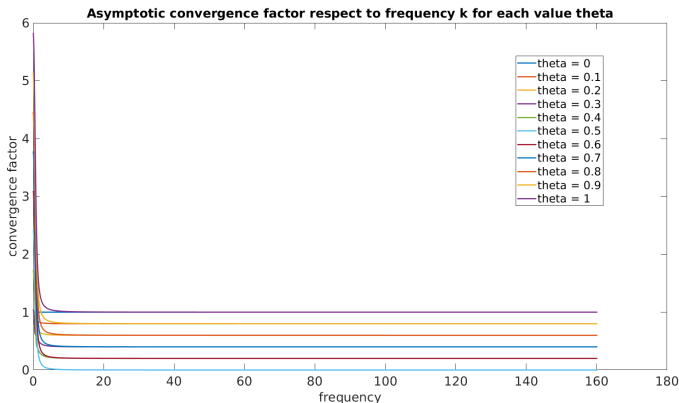


Figure: Asymptotic convergence factor of Schwarz algorithm respect to frequency k for some value θ

Equioscillation

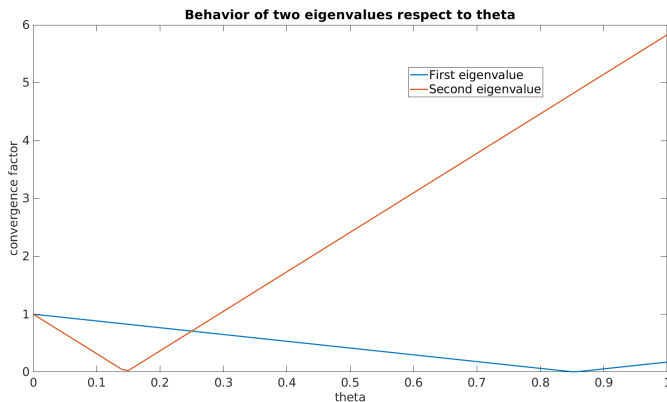


Figure: Two eigenvalue respect to theta when frequency $k = 0$

Optimal theta

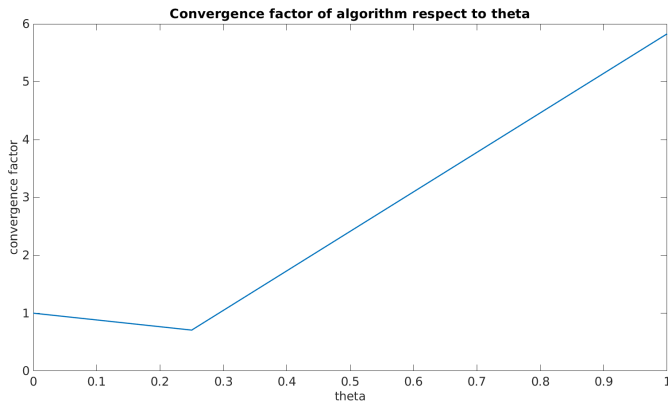


Figure: Convergence factor of Schwarz algorithm respect to θ

Schwarz with $\theta = 0.25$ when T large

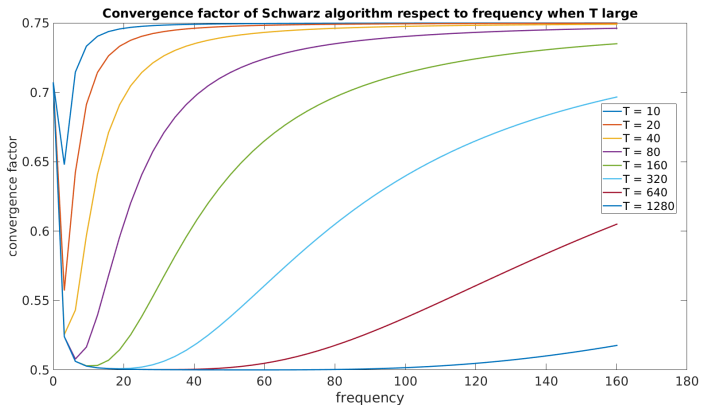


Figure: Convergence factor of Schwarz algorithm respect to frequency k when T large

Schwarz algorithm with γ'

There is a possibility to optimize the Schwarz algorithm with parameter γ' in transmission condition on the domain $\Omega \times (0, T_1)$.

$$\begin{cases} \lambda_1^{n+1}(x, T_1) = \lambda_2^n(x, T_1), & x \in \Omega, \\ \lambda_{1,t}^{n+1}(x, T_1) - \gamma' y_1^{n+1}(x, T_1) = \lambda_{2,t}^n(x, T_1) - \gamma' y_2^n(x, T_1), & x \in \Omega \end{cases}$$

Theorem (7)

For $T < \tilde{T}_0$, the Schwarz algorithm converges with γ' close to 0.

Conjecture (2)

- There exists a T_0^* such that for $T < T_0^*$, the Schwarz algorithm converges for some values of γ' and there is an optimal γ'_{opt} .
- When T large enough, there is no value of γ' such that the Schwarz algorithm converges.

Numerical result with Schwarz and γ' when $T = 2$

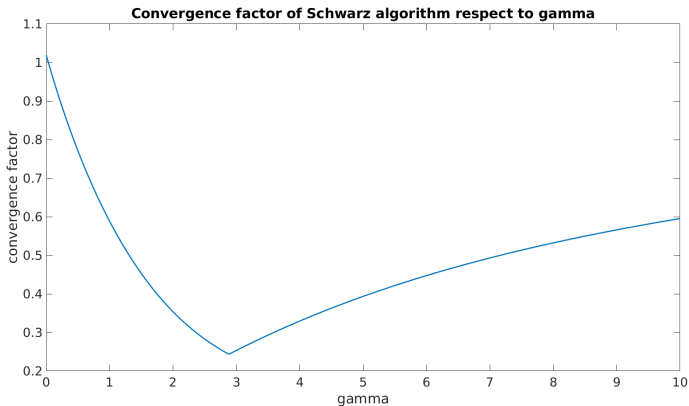


Figure: Convergence factor of Schwarz algorithm respect to γ when $T = 2$

Numerical result with Schwarz and γ' when $T = 4$

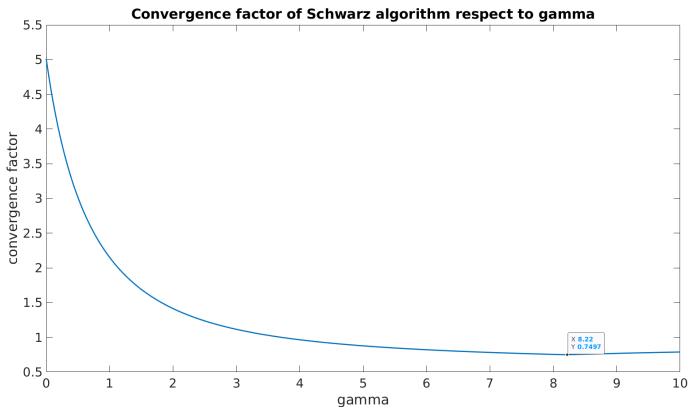


Figure: Convergence factor of Schwarz algorithm respect to γ when $T = 4$

DIRICHLET NEUMANN METHOD IN TIME

Dirichlet Neumann algorithm

Different than Schwarz algorithm, the Dirichlet Neumann algorithm changes the transmission condition.

$$\begin{cases} y_{1,tt}^{n+1} - y_{1,xx}^{n+1} - \lambda_1^{n+1} = 0, & (x, t) \in \Omega \times (0, T_1), \\ \lambda_{1,tt}^{n+1} - \lambda_{1,xx}^{n+1} + y_1^{n+1} = \hat{y}, & (x, t) \in \Omega \times (0, T_1), \\ y_1^{n+1}(x, 0) = y_{1,t}^{n+1}(x, 0) = 0, & x \in \Omega, \\ y_1^{n+1}(x, T_1) = y_2^n(x, T_1), & x \in \Omega, \\ \lambda_1^{n+1}(x, T_1) = \lambda_2^n(x, T_1), & x \in \Omega, \end{cases}$$

$$\begin{cases} y_{2,tt}^{n+1} - y_{2,xx}^{n+1} - \lambda_2^{n+1} = 0, & (x, t) \in \Omega \times (T_1, T), \\ \lambda_{2,tt}^{n+1} - \lambda_{2,xx}^{n+1} + y_2^{n+1} = \hat{y}, & (x, t) \in \Omega \times (T_1, T), \\ \lambda_2^{n+1}(x, T) = \lambda_{2,t}^{n+1}(x, T) = 0, & x \in \Omega \\ y_{2,t}^{n+1}(x, T_1) = y_{1,t}^{n+1}(x, T_1), & x \in \Omega, \\ \lambda_{2,t}^{n+1}(x, T_1) = \lambda_{1,t}^{n+1}(x, T_1), & x \in \Omega. \end{cases}$$

Dirichlet Neumann algorithm on finite domain

- With relaxation parameter θ , we change the transmission condition

$$\begin{cases} y_1^{n+1}(x, T_1) = (1 - \theta)y_1^n(x, T_1) + \theta y_2^n(x, T - T_1), \\ \lambda_1^{n+1}(x, T_1) = (1 - \theta)\lambda_1^n(x, T_1) + \theta \lambda_2^n(x, T - T_1). \end{cases}$$

- convergence factor is spectral radius of

$$M_{DN,\theta}(k) = I - \Theta(I - M_{DN}(k)).$$

Divergence of Dirichlet Neumann algorithm

Theorem (8)

Dirichlet Neumann algorithm diverges when $T = 1$

With asymptotic computation for k large, the eigenvalue of matrix M are $\mu_1 \sim -\frac{4k^2}{T_1^2} + O(k)$ and $\mu_2 \sim O(k)$.

Numerical illustration

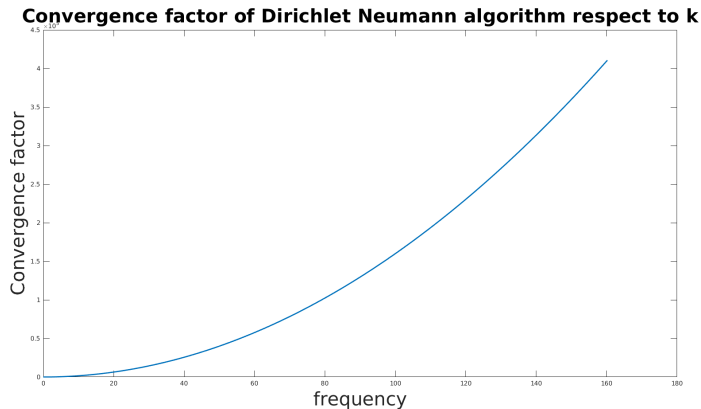
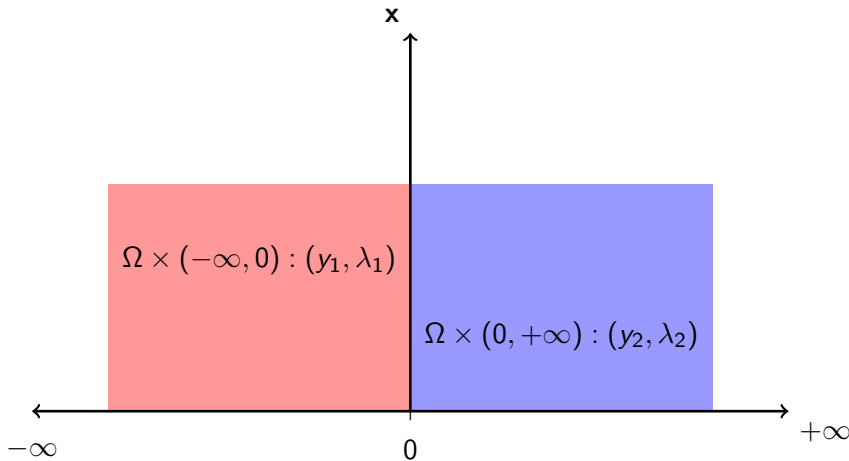


Figure: Dirichlet Neuman algorithm with frequency k

Dirichlet Neumann algorithm over infinite domain



Dirichlet Neumann algorithm over infinite domain

$$\begin{cases} y_{1,tt}^{n+1} - y_{1,xx}^{n+1} - \lambda_1^{n+1} = 0, & (x, t) \in \Omega \times (-\infty, 0), \\ \lambda_{1,tt}^{n+1} - \lambda_{1,xx}^{n+1} + y_1^{n+1} = \hat{y}, & (x, t) \in \Omega \times (-\infty, 0), \\ \lim_{t \rightarrow -\infty} y_1^{n+1}(x, t) = 0 & x \in \Omega, \\ y_1^{n+1}(x, 0) = y_2^n(x, 0), & x \in \Omega, \\ \lambda_1^{n+1}(x, 0) = \lambda_2^n(x, 0), & x \in \Omega, \end{cases}$$

$$\begin{cases} y_{2,tt}^{n+1} - y_{2,xx}^{n+1} - \lambda_2^{n+1} = 0, & (x, t) \in \Omega \times (0, +\infty), \\ \lambda_{2,tt}^{n+1} - \lambda_{2,xx}^{n+1} + y_2^{n+1} = \hat{y}, & (x, t) \in \Omega \times (0, +\infty), \\ \lim_{t \rightarrow +\infty} \lambda_2^{n+1}(x, t) = 0, \\ y_{2,t}^{n+1}(x, 0) = y_{1,t}^{n+1}(x, 0), & x \in \Omega, \\ \lambda_{2,t}^{n+1}(x, 0) = \lambda_{1,t}^{n+1}(x, 0), & x \in \Omega. \end{cases}$$

Dirichlet Neumann algorithm on infinite domain

Theorem (9)

For any $\theta \in (0, 1)$, the algorithm converges. The optimal $\theta_{opt} = \frac{1}{2}$ for which the convergence factor vanishes.

Compute the matrix $M = -I$ explicitly and the eigenvalue is equal to -1 . With relaxation θ , we have the eigenvalues are $1 - 2\theta$.

Numerical result

Convergence factor of Dirichlet Neumann algorithm over infinite time domain respect to theta

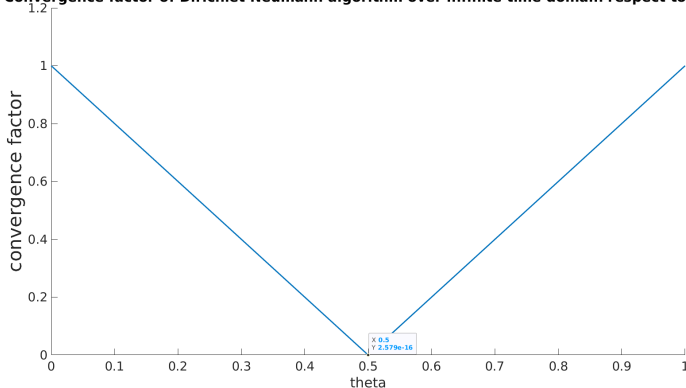


Figure: Convergence factor of Dirichlet Neumann algorithm over infinite time interval respect to theta

Dirichlet Neumann algorithm over finite domain

$$\begin{cases} y_{1,tt}^{n+1} - y_{1,xx}^{n+1} - \lambda_1^{n+1} = 0, & (x, t) \in \Omega \times (-\frac{T}{2}, 0), \\ \lambda_{1,tt}^{n+1} - \lambda_{1,xx}^{n+1} + y_1^{n+1} = \hat{y}, & (x, t) \in \Omega \times (-\frac{T}{2}, 0), \\ y_1^{n+1}(x, -\frac{T}{2}) = y_{1,t}^{n+1}(x, -\frac{T}{2}) = 0, & x \in \Omega, \\ y_1^{n+1}(x, 0) = y_2^n(x, 0), & x \in \Omega, \\ \lambda_1^{n+1}(x, 0) = \lambda_2^n(x, 0), & x \in \Omega, \end{cases}$$

$$\begin{cases} y_{2,tt}^{n+1} - y_{2,xx}^{n+1} - \lambda_2^{n+1} = 0, & (x, t) \in \Omega \times (0, \frac{T}{2}), \\ \lambda_{2,tt}^{n+1} - \lambda_{2,xx}^{n+1} + y_2^{n+1} = \hat{y}, & (x, t) \in \Omega \times (0, \frac{T}{2}), \\ \lambda_2^{n+1}(x, \frac{T}{2}) = 0, \lambda_{2,t}^{n+1}(x, \frac{T}{2}) = 0, & x \in \Omega \\ y_{2,t}^{n+1}(x, 0) = y_{1,t}^{n+1}(x, 0), & x \in \Omega, \\ \lambda_{2,t}^{n+1}(x, 0) = \lambda_{1,t}^{n+1}(x, 0), & x \in \Omega. \end{cases}$$

Dirichlet Neumann algorithm on finite domain

Theorem (10)

Given any relaxation parameter $\theta \in (0, 1)$, there exists a $T(\theta)$ such that for any $T \geq T(\theta)$ the Dirichlet Neumann algorithm converges over the finite domain $[-\frac{T}{2}, \frac{T}{2}] \times \Omega$. Particularly, convergence factor of algorithm depending on θ tends to $1 - 2\theta$ when $T \rightarrow +\infty$ and it converges to zero if $\theta = 0.5$.

Sketch of proof:

- Make an asymptotic expansion for variable T .
- Eigenvalue of M asymptotically $\lambda_M \sim -1 + O(\exp(-2\nu_2 T_1))$.
- With relaxation parameter $\theta \in (0, 1)$,
 $\lambda_{M_\theta} \sim 1 - 2\theta + O(\exp(-2\nu_2 T_1))$.

Numerical result

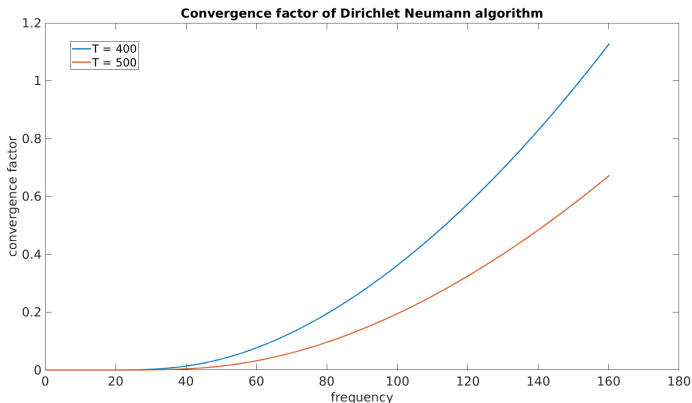


Figure: Convergence factor of Dirichlet Neumann algorithm when $\theta = 0.5$

Reference



Martin J.Gander, Laurence Halpern and Frédéric Nataf, Optimal Schwarz Waveform relaxation for the one dimensional wave equation.



P.G.Ciarlet, Introduction à l'analyse numérique matricielle et à l'optimisation.



Matthias Heinkenschloss, A time-domain decomposition iterative method for the solution of distributed linear quadratic optimal control problems, 2003.



Richard Krug, Günter Leugering, Alexander Martin, Martin Schmidt, Dieter Weninger, Time-Domain Decomposition for Optimal Control Problems Governed by Semilinear Hyperbolic Systems, 2021.



Felix Kwok, On the Time-domain Decomposition of Parabolic Optimal Control Problems, 2017.

Reference

-  Martin J.Gander and Felix Kwok, Schwarz Methods for the Time-Parallel Solution of Parabolic Control Problems, 2016.
-  Andrew T. Barker and Martin Stoll, Domain decomposition in time for PDE-constrained optimization, 2015.
-  J.E.Lagnese and G.Leugering, Time domain decomposition of optimal control problems for the wave equation, 2002.
-  Stephen Boyd and Lieven Vandenberghe, Convex Optimization, Cambridge University Press, 2004.
-  Yafei Sun, Shu-Lin Wu and Yingxiang Xu, A Parallel-in-Time Implementation of the Numerov Method for Wave Equations, 2021.

Reference



Martin J. Gander and Yongxiang Liu, Is there more than one Dirichlet-Neumann algorithm for the biharmonic problem, 2021.



Martin J. Gander and Yongxiang Liu, On the Definition of Dirichlet and Neumann Conditions for the Biharmonic Equation and Its Impact on Associated Schwarz Methods, 2016.



Bankim Chandra Mandal, PhD thesis, Convergence Analysis of Substructuring Waveform Relaxation Methods for Space-time Problems and Their Application to Optimal Control Problems, 2014.



B. C. Mandal, A Time-Dependent Dirichlet-Neumann Method for the Heat Equation, Domain Decomposition in Science and Engineering XXI, LNCSE, Vol. 98, Springer-Verlag 2014, pp. 467-475.

Reference



Martin J. Gander, Felix Kwok and Bankim C.Mandal, Dirichlet-Neumann and Neumann-Neumann waveform relaxation algorithms for parabolic problems, 2016.



M. J. Gander, Felix Kwok, B. C. Mandal, Dirichlet-Neumann and Neumann-Neumann Waveform Relaxation for the Wave Equation, 2015.



R.T.Rockafellar, Convex analysis, Princeton University Press, 1970.



M. J. Gander, Felix Kwok, B. C. Mandal, Dirichlet-Neumann Waveform Relaxation Algorithm for the Heat and Wave Equations in Multiple subdomains, 2020.